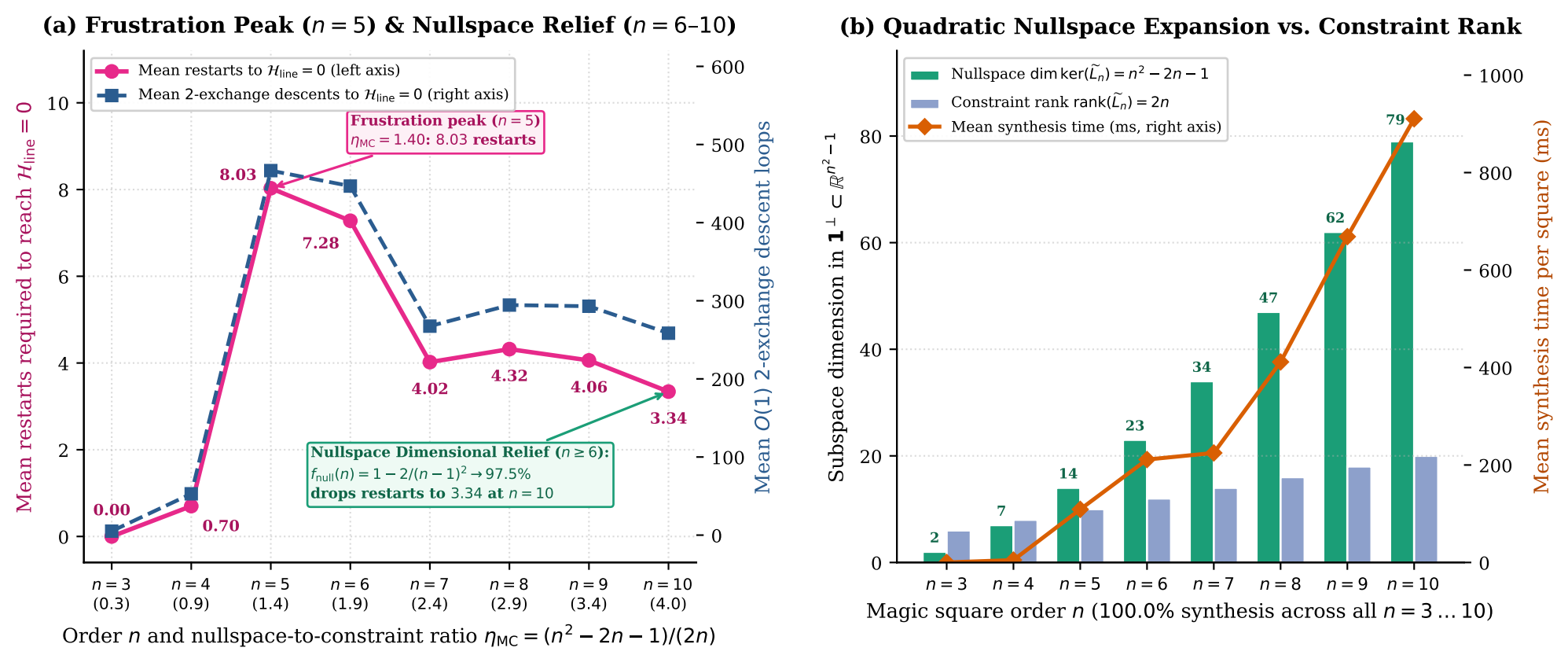

(a) Frustration Peak ($n = 5$) & Nullspace Relief ($n = 6$–10)
Mean restarts to $\mathcal{H}_{\text{line}} = 0$ (left axis)
Mean 2-exchange descents to $\mathcal{H}_{\text{line}} = 0$ (right axis)
Frustration peak ($n = 5$)
$\eta_{\text{MC}} = 1.40$: 8.03 restarts
Nullspace Dimensional Relief ($n \geq 6$):
$f_{\text{null}}(n) = 1 - 2/(n-1)^2 \to 97.5\%$
drops restarts to 3.34 at $n = 10$
Mean restarts required to reach $\mathcal{H}_{\text{line}} = 0$
Mean $O(1)$ 2-exchange descent loops
0.00
0.70
8.03
7.28
4.02
4.32
4.06
3.34
$n = 3$ (0.3)
$n = 4$ (0.9)
$n = 5$ (1.4)
$n = 6$ (1.9)
$n = 7$ (2.4)
$n = 8$ (2.9)
$n = 9$ (3.4)
$n = 10$ (4.0)
Order n and nullspace-to-constraint ratio $\eta_{\text{MC}} = (n^2 - 2n - 1)/(2n)$
(b) Quadratic Nullspace Expansion vs. Constraint Rank
Nullspace $\dim \ker(\widetilde{L}_n) = n^2 - 2n - 1$
Constraint rank $\text{rank}(\widetilde{L}_n) = 2n$
Mean synthesis time (ms, right axis)
Subspace dimension in $\mathbf{1}^{\perp} \subset \mathbb{R}^{n^2-1}$
Mean synthesis time per square (ms)
2
7
14
23
34
47
62
79
Magic square order n (100.0% synthesis across all $n = 3 \ldots 10$)